

EV BATTERY HEALTH MONITORING AND VISUALIZATION SYSTEM

A PROJECT REPORT

**BACHELOR OF TECHNOLOGY
IN
(ELECTRICAL AND ELECTRONICS)**

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ABSTRACT

Electric vehicles (EV) are becoming an increasingly large component of a worldwide transition to a cleaner and more sustainable transport model, thus electric vehicles (EVs) represent one of the emerging solutions to environmental pollution and fossil fuel scarcity. The EV battery is the energy storage device, and because of this, all the performance, cost and safety of a battery-powered EV depend on how the battery is managed. An EV battery's reliability is critical for enhancing the driving range of the vehicle, increasing the vehicle's efficiency, and providing stability to the entire vehicle system.

A major issue impacting EV batteries is the gradual deterioration in their performance over time. All batteries degrade, which means that battery capacity is reduced over time as they age. Aging typically results from several different factors, including environmental operating temperature, charging/discharging patterns, depth of discharge and number of cycles used. Specifically, if a battery continues to operate at elevated temperatures, the chemical reactions inside the battery can increase, increasing the rate at which it degrades (ages). In addition, frequent charging and deep discharging of a battery will also decrease the battery's ability to store energy and subsequently reduce the battery's performance, impacting the vehicle's range and efficiency. Therefore, accurately monitoring an EV battery's health and ongoing predictive capabilities is a crucial task.

This project will assume the responsibility to develop a smart, data-driven monitoring and predictive capability for an EV battery to determine its state of charge (SOC), temperature, cycle count, and operational time, as well as other critical parameters. The parameters derived from this simulation model, which is constructed using MATLAB/Simulink, are used to create a simulated battery that represents the actual performance of lithium-ion batteries in a variety of situations through simulation. The simulation of various battery scenarios enables the production of valid data sets for post-simulation analysis. Battery degradation due to aging is represented as an exponential decline in usable capacity, as this is the typical pattern for lithium-ion batteries over time. The State of Health (SOH) of the battery indicates the current usable capacity relative to the original, and the State of Health (SOH) values are used to categorize the battery according to its current condition (Good, Warning, or End of Life [EOL]).

In addition to the health assessment, the Remaining Useful Life (RUL) of the battery can also be predicted. This assists in anticipating when a battery will likely be unable to provide sufficient power to support its intended application, thus allowing for an appropriate advance notice to be sent for replacement prior to the actual discharge. The simulation results also produce a structured data set that can be saved in various formats such as Excel or CSV files. This data can be leveraged for machine learning and thus facilitate the enhancement of the predictive capabilities of future models.

In conclusion, the overall project demonstrates a solution for enhancing the battery monitoring and management of electric vehicles by leveraging the integration of simulation, data analysis, and visualization to improve battery lifespan and decrease battery cost through the advancement of sustainable transportation.

INTRODUCTION

In light of global trends towards more sustainable energy sources and environmentally conscious technologies, the emergence of electric powered vehicles (EV) is a major factor in the reduction of carbon emissions and fossil fuel dependence; therefore, several countries/areas around the world are creating long-term plans that incorporate EV use as a component of their overall environmental strategy. Even with significant advances in the development of EV technology over the years, the actual performance and dependability of EVs continues to be largely related to the current condition of the vehicle's batteries. This is because the battery is the primary source of energy for an EV, and its performance directly impacts vehicle efficiency, safety, and distance travelled. A main area of concern in relation to EV batteries is the natural degradation of battery systems over time. This process is due to a combination of internal chemical reactions within the battery and external environmental factors, or operating conditions. For instance, as battery temperatures rise, the rate of the internal chemical reactions that occur within the battery are accelerated, which will accelerate the battery's age.. Over time, as the stress on the battery continues to accumulate, the battery's ability to hold and supply energy will be compromised, which can ultimately affect the vehicle's performance and reliability.

The primary function of traditional battery management systems (BMS) is to provide real-time monitoring of the basic operating parameters of the battery, such as voltage, current, and temperature. Although these parameters are essential for safe operation of the battery, they do not provide sufficient information regarding the overall health of the battery in the long term or about the potential future state of the battery. Due to this fact, it is often impossible for users to reliably determine when a battery will substantially degrade or when a battery should be replaced. This project proposes a data-driven method of monitoring and forecasting battery health. The system leverages mathematical modelling and data analysis techniques to provide a better understanding of how batteries perform. By applying an exponential degradation model, the system is able to simulate how batteries deteriorate over time and use this simulation to make estimations of key performance metrics, including state of health (SOH).

The system has been designed to enable the use of predictive analytics/machine learning so that users can leverage the growing dataset to improve the accuracy of future predictions. The input parameters to the system include SOC (state of charge), temperature, cycle count, operational time, etc., which the battery condition will be analysed in real-time. Using these input parameters, the battery degradation trend will be generated, and a forecast of remaining useful life (RUL) will be made. In order to enhance the user's experience with the use of this system, a real-time dashboard has been developed to help facilitate visualisation of the key information provided by the system, e.g. SOC, SOH, temperature variances, and predicted lifespan. The dashboard provides graphs and indicators to assist users in understanding the state of health and monitoring the state of the battery.

The goal of this project is to develop a sophisticated and intuitive solution for the monitoring of battery health in electric vehicles. By coupling analytical models with data-driven analytics, the solution helps optimise battery management and provides support for the predictive maintenance of batteries and the continued efficient growth and development of EV technology.

OBJECTIVES

The main objective of this project is to develop a smart and efficient system to monitor electric vehicle (EV) battery parameters in real time. The system will focus on tracking key parameters including State of charge (SOC), temperature, cycle count and the time taken to charge and discharge the battery as these parameters all have an impact on the performance of the battery. The system will then use this information combined with the appropriate degradation models to estimate the State of Health (SOH) of the battery giving the user an accurate view of the battery condition. The system will classify the battery according to the SOH values into various categories such as good, warning and end-of-life to better allow the user to understand the battery status and condition.

The second objective of this project is to use predictive models to predict the remaining useful life (RUL) of the battery to assist in planning routine maintenance for the battery and avoiding sudden failure. To create a more sophisticated and fully prepared for the future system, the project will create a structured dataset that can be utilized for machine learning and further predictive analysis. Finally, the project will create a user-friendly dashboard that will allow for real-time viewing of battery performance through graphs and indicators, making it easier to use and applicable to real-world EV deployments.

METHODOLOGY

The methodology of this project is a structured and data-driven approach to effectively monitor and predict the health of Electric Vehicle (EV) batteries. In this stage of development, either by gathering or simulating important battery characteristics, such as the State of Charge (SOC), average & maximum temperature, Cycle Count, Charging/ Discharge Times, among others, it will be essential for the Environment to determine how well a particular battery performs and the speed at which a specific battery degrades. Instead of utilizing only fixed values, the Environment will utilize both a stable set of conditions and any real-time variations in the Variables that influence battery performance. Utilizing both Static & Dynamic Conditions provides the User with a much more accurate representation of battery performance. Together, Static Values, Dynamic Conditions, and the Regular Reference Conditions will be created into an Organized Data Set. This Data Set will be beneficial in determining the actual performance of a battery and will prepare the Environment for future machine learning applications.

The degradation of Lithium-Ion Batteries will be modelled using an Exponential Degradation Model. The Exponential Degradation Model closely mimics how the Lithium-Ion Battery degrades in Capacity over time. An initial application of a Base Degradation Factor represents a gradual degradation. The Base Degradation Factor will be supplemented by the application of Additional Stress Factors due to Increasing Temperature Levels and Usage Cycle Limits, thereby offering the User's Battery Performance as a more accurate and practical simulation of what the User's Battery would actually see in a Real-World Environment. Using the calculated Degradation Values, the State of Health (SOH) of the Battery is determined and is a direct representation of what is left of the battery's Capacity compared to the Original Condition of the Battery.

Once the SOH is calculated, the Environment will categorize the Battery into Health Categories and present the Results in the Category Format. For instance, if the SOH is Greater Than 90%, the Battery will be classified as Excellent; if the SOH is Between 75% and 90%, the Battery will be classified as a Moderate or Good Battery. If the SOH is Less Than 75%, the Battery is considered Poor and may either require Monitoring or Replacement. Categorizing the SOH makes it easier for Users to determine the Battery's Status without requiring advanced technical knowledge.

In Addition to SOH, the Environment will also calculate the Remaining Useful Life (RUL) of the Battery based on comparing the Total Expected Life Cycle of the Battery with the Number of Cycles that have been performed. The RUL Calculation will take into consideration that different battery types (e.g., LFP, NMC, NCA) have differing Limitations on Cycle Counts. The RUL Calculation will prove to be of great assistance to Users in their Maintenance Planning and help Users avoid unanticipated Failure Events.

Last, the Overall Environment's DBM will continuously update the Battery Data in Realtime, usually at intervals of 0.5 Seconds, via a Back-end Process. A User-Friendly Dashboard is tied to the Environment to display various parameters, such as SOC, SOH, Temperature Trends, and RUL, using indicators and Graphs to provide a Comprehensive Overview of the Battery's Performance within seconds at a glance. The Methodology combines Simulation, Data Analysis, and Visualization together to provide an Accessible, Reliable, and Practical Method for Monitoring Battery Performance in Electric Vehicles.

DATASET GENERATION USING MATLAB SIMULINK

1) CHEMISTRY SELECTOR AND CHEMISTRY LIMITS BLOCK

There is an initial block in this design called the chemistry selector, which is responsible for selecting the battery chemistry to evaluate. Possible chemistries for evaluation include Lithium Iron Phosphate (LFP), Nickel Manganese Cobalt (NMC) and Nickel Cobalt Aluminium (NCA). Each of these chemistries offers different behaviours and characteristics, performance and degradation. The next block in the chain is a limits block that establishes the safety and recommended operating limits for a specific battery chemistry, i.e., what the maximum and minimum state of charge, what the acceptable temperature range is and what the limits are for total cycle life will be. The limits established in this block are critical because they serve as reference points against which the operation of the battery can be measured. By defining these limits before any calculations or simulations take place, it allows for future calculations and simulations to be performed within the limits of real-world battery specifications, thereby providing accurate health estimates and performance monitoring.

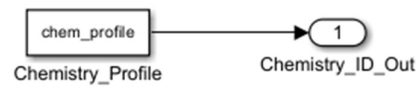


Fig 2: Internal System of Chemistry Selector Block

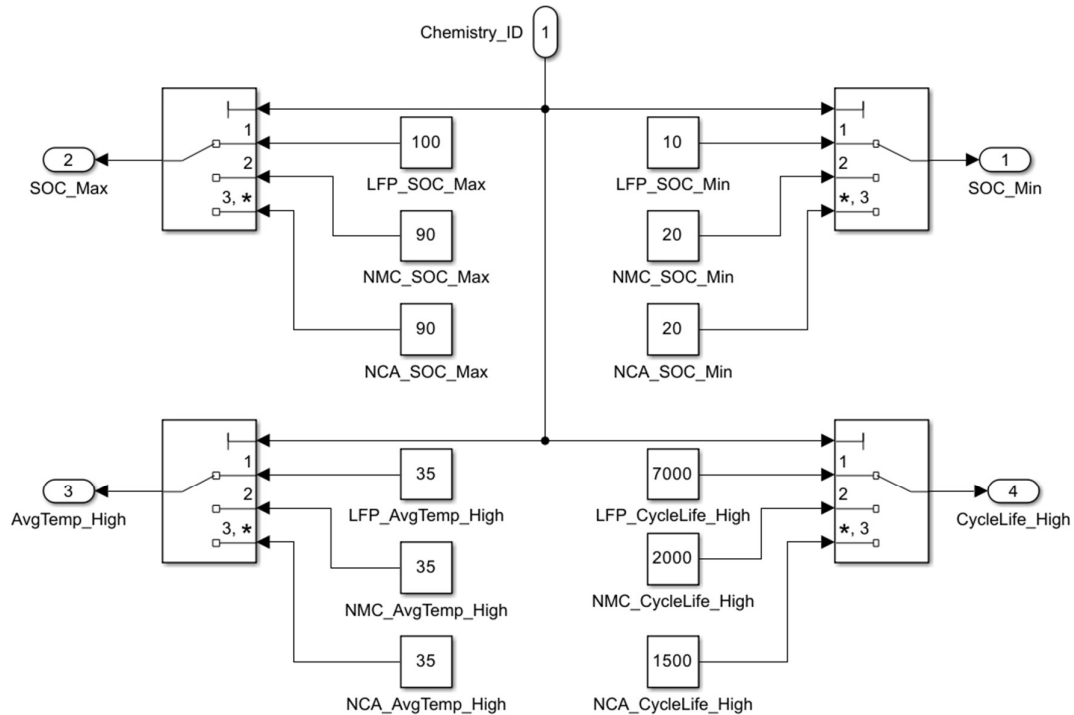


Fig 3: Internal System of Chemistry Limits Block

2) SOC GENERATOR AND SOC STRESS BLOCK

The purpose of the SOC generator block is to provide a simulation of how the state of charge of a battery will change over time due to charging and discharging cycles. The SOC generator simulates what happens to a battery in real life and produces variable charge levels over time. The SOC values produced from the SOC generator are sent to the SOC stress block, where these SOC values will be compared with the limits established in the chemistry limits block to identify when the battery is operating at extreme levels and will be stressed due to high or low SOC's, increasing degradation in the battery over time. The SOC Stress Block provides information on how much damage the SOC will cause on crossing limits by producing a stress value that indicates the degree of damage caused by the SOC Operating Region. As the SOC stress is continuously monitored, trends can be identified that allow the system to develop methods to extend the life of the battery by controlling its operation and/or predicting its usage patterns.

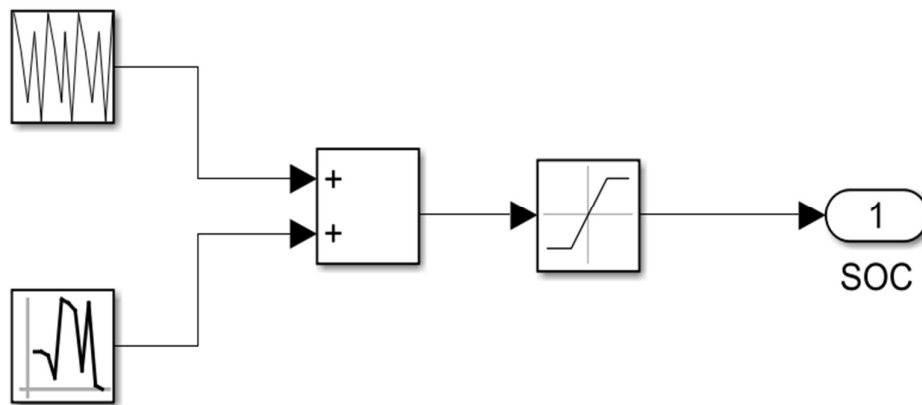


Fig 4: Internal System of SOC Generator Block

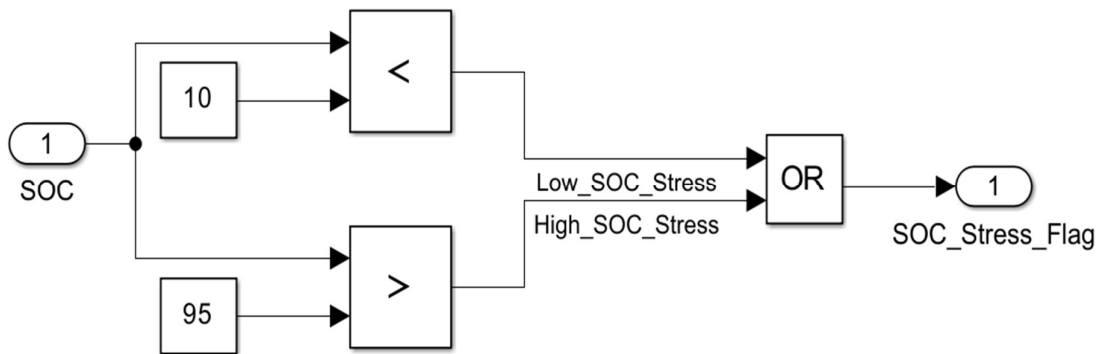


Fig 5: Internal System of SOC Stress Block

3) TEMPERATURE GENERATOR AND TEMPERATURE STRESS BLOCK

The Temperature Generator Block simulates the thermal environment of a Battery in the Operating Region. The Temperature Generator produces both a Mean Temperature and a Peak Temperatures to represent environmental heat and heat generated internally by the Battery during operation. The Mean and Peak Temperature data is used by the Temperature Stress Block, which calculates thermal stress on the battery cells when the Mean and Peak Temperatures exceed allowable limits and/or when the Mean and Peak Temperatures change significantly.

High Mean and Peak Temperatures will accelerate Battery chemical degradation and reduce the Battery's useful life. The Temperature Stress Block provides a quantitative measure of the degree of thermal impact on the battery and provides the data needed to estimate the battery's state of health. Thermal stress monitoring is critical to the safety, performance, and longevity of Batteries in Electric Vehicle Systems, since they can be impacted severely by the thermal environment.

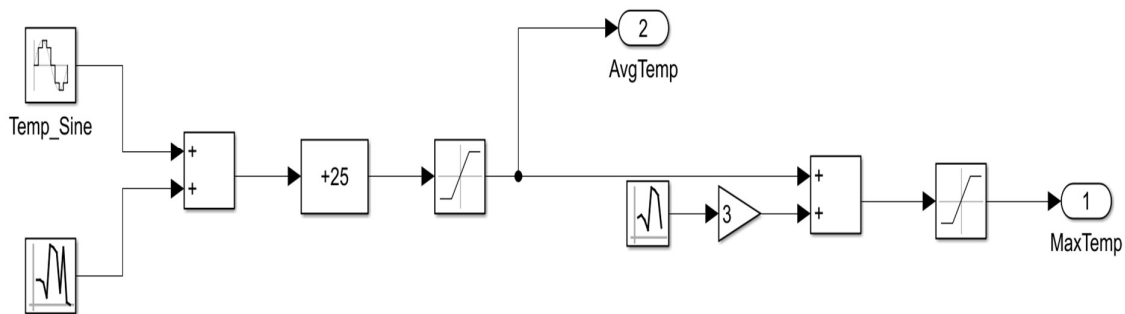


Fig 6: Internal System of Temperature Generator Block

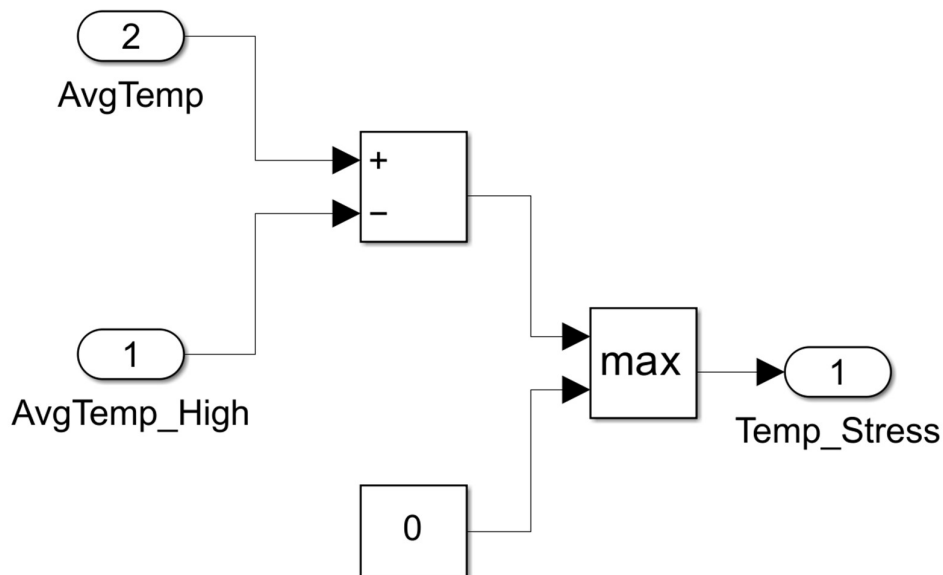


Fig 7: Internal System of Temperature Stress Block

4) CYCLE GENERATOR BLOCK

The Cycle Generator Block simulates the total number of Charge and Discharge Cycles a Battery experiences during its life. The battery lifecycle encompasses its entire cycle of charging and discharging, as well as its ability to maintain functionality over time. The battery's ability to hold charge decreases with each complete cycle, The battery's cycle count reflects its history of usage and provides an estimate of its remaining lifespan. This data is used to create accurate usage forecasts for battery health. In turn, the cycle count also contains valuable information about current battery capacity degradation. Accurate tracking of cycle counts supports the system to produce accurate predictions and projections regarding the degradation of the battery over time and forecasts regarding how long the battery will remain operational. The cycle count is therefore key to accurately reflecting real-world battery usage and degradation behaviour.

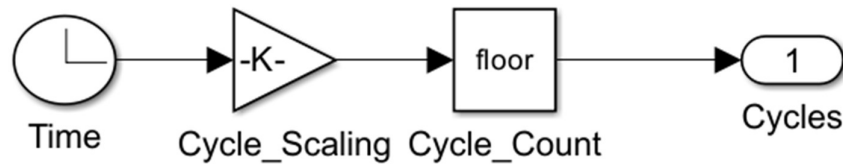


Fig 8: Internal System of Clock Generator Block

5) THE SOH ESTIMATOR BLOCK

As the most reliant and scientifically based component of the tracking system, the state of health estimator block uses the SOC stress, temperature stress, cycle count, and chemistry data to produce a reliable mathematical output that estimates the condition of the battery. By developing a unique degradation model that is based on the historical degradation patterns of batteries, the state of health estimator block will provide an estimate of the degradation of the battery from its original capacity (e.g., based on an exponential curve). Therefore, the output of the state of health estimator block is a percentage of battery capacity representing the condition of the battery and its remaining capacity. The higher the SOH value, the healthier the battery; the lower the SOH value, the more severely degraded the battery. This output allows the user to analyse battery performance so that future repairs or replacements may be planned. The final phase of battery health monitoring combines all historical data and all stressors (pulls, pushes, pulls) into one combined metric - i.e., it takes into account all factors affecting battery health at once.

The formula for SOH Estimator is given by

$$SOH = 100 - \text{Degradation value}$$

$$\text{Degradation Value} = \text{Aging Constant} + \text{Cycle Degradation}$$

$$+ \text{Temperature Degradation} + \text{SOC Degradation}$$

$$\text{Cycle}_{\text{Degradation}} = \Delta \text{cycles} \times \text{constant}(a_0)$$

$$\text{Temperature}_{\text{Degradation}} = \text{Temperature}_{\text{Stress}} \times \text{constant}(b_0 = 0.3)$$

$$SOC_{Degradation} = SOC_{Stress} \times constant(c_0 = 2)$$

The constant value (a_0) for various chemistries of Batteries are given in the table below

PARAMETER	LFP	NMC	NCA
(a_0)	0.002	0.004	0.006

Table 1: Cycle Degradation constant

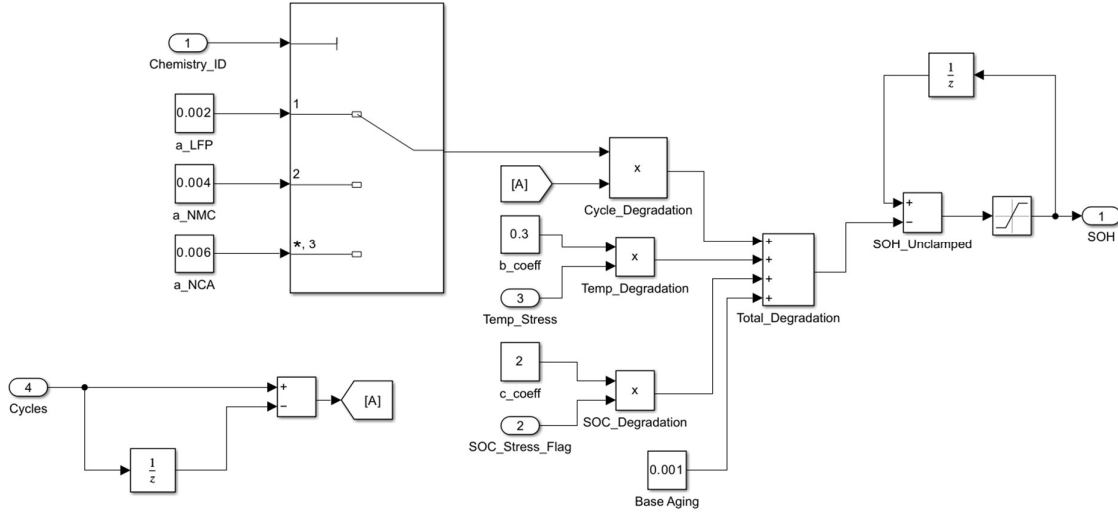


Fig 9: Internal System of SOH Estimator Block

6) RUL ESTIMATOR

The remaining useful life (RUL) estimator creates a predictive estimate of the remaining useable life of a given battery once it reaches the end of its effective lifetime. The primary inputs to this block are the battery state of health, the allowable total cycle life for the given battery chemistry and the total cycle count to date. Using information about the current day use as compared to the maximum allowed cycle capacity, RUL estimates how many cycles, or how long, are left before that battery likely becomes ineffective or completely unusable. By providing an estimate of RUL to users this will allow for advanced scheduling of future maintenance periods and will minimize chances of unexpected battery expirations. In EV applications, where reliability is paramount, having this knowledge will allow for proficient scheduling of replacement and/or maintenance activities based upon improved estimates of RUL. By providing this accurate prediction of usable life remaining, users benefit from improved battery performances and reliability.

- The SOH Factor is given by

$$SOH_{factor} = \frac{SOH}{100}$$

- Remaining Cycle Factor is given by

$$RemainingCycleFactor = 1 - \frac{Cycles}{CycleLife_{High}}$$

c. Temperature Factor is given by

$$TempFactor = 1 - Temp_{Stress}$$

d. Nominal Life (based on Chemistry)

e. Final RUL Equation

$$RUL = NominalLife \cdot SOH_{factor} \cdot RemainingCycleFactor \cdot TempFactor$$

The Nominal Life of Various Chemistries of Batteries are given in the below table

PARAMETER	LFP	NMC	NCA
Nominal Life	0.002	0.004	0.006

Table 2: Nominal Life of the Batteries

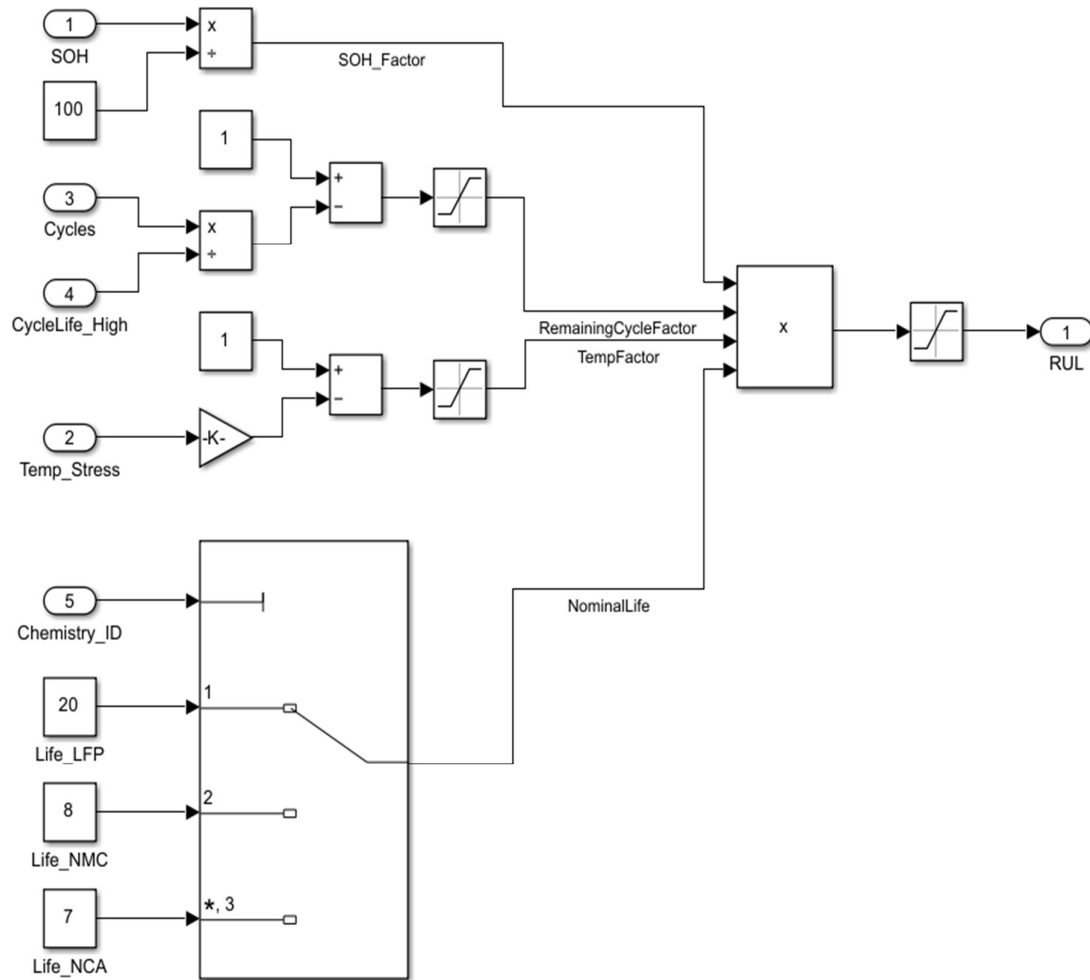


Fig 10: Internal System of RUL Estimator Block

7) HEALTH ESTIMATOR BLOCK

The health estimator converts the battery state of health numeric value into "simple" classification terms (excellent, good and poor). For example, instead of stating the state of health of the battery in percent (such as 30 percent good, etc.), the health estimator classifies the battery state into easy to understand classification levels of excellent, good or poor. Thus, providing an easy means for end users to quickly identify the current battery health and status without technical battery knowledge. Through classification, for example, a battery that had high state of health would be classified as excellent; on the other hand, a battery with lower value of state of health would be classified as poor. Classification aids in usability and decision-making and allows for ease of interpretation of battery condition by operators, engineers or non-technical users. By providing a straightforward means of translating complicated data into more straightforward terminology, Blocks 8 through 10 of our Technology provide greater user experience through clarity. Operators are also able to better communicate critical battery data without the need for extensive training.

The Health Status ranges are given in the below table

PARAMETER	GOOD	WARNING	END OF LIFE
Health Status	More than 80	Between 70 and 80	Less than 70

Table 2: Nominal Life of the Batteries

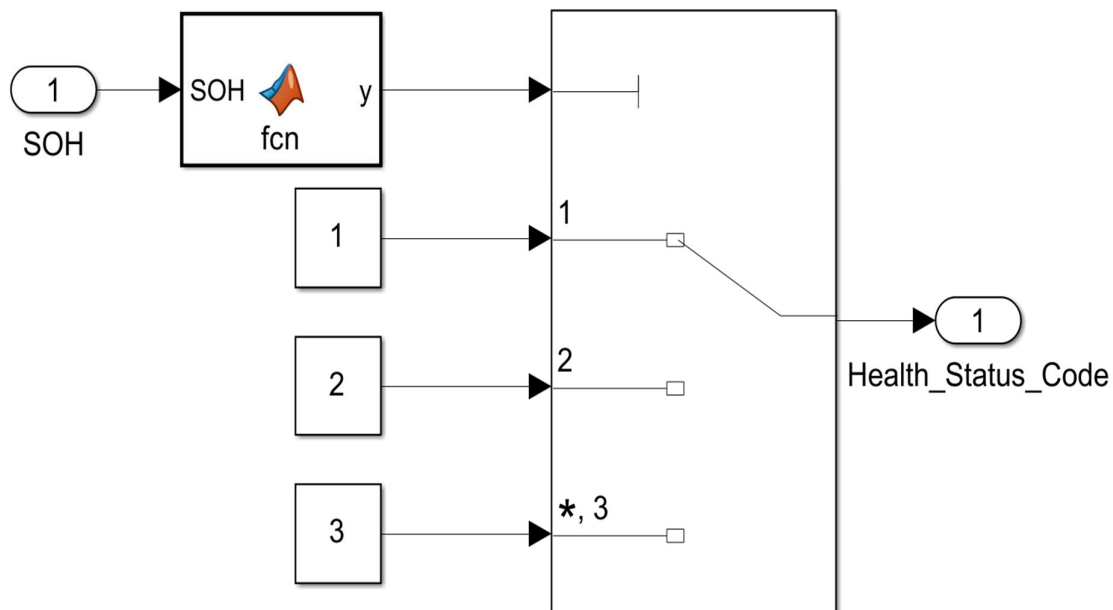


Fig 11: Internal System of Health Estimator Block

8) ALARM GENERATOR

Alarms are used as a primary safety feature for most systems. Alarm Generator blocks continuously monitors the battery through monitoring of critical parameters such as State of Health (SOH), State of Charge (SOC) and Temperature; should any monitored data exceed predetermined limits in regards to the conditions being monitored, an alarm signal is then generated. An example would include if the Temperature of the battery were to exceed its established limits or the SOH dropped to an unacceptable level, an alarm would be generated. This alerts operators of the battery about possible risks associated with the abnormal conditions experienced by the battery. The Alarm Generator blocks can reduce the risk of battery failure, damage, and enhance overall safety. In addition, Alarm Generator blocks provide operators with information needed to make timely decisions regarding battery performance. Therefore, the Alarm Generator block is essential for ensuring the reliability of the system and protects both the battery and the surrounding environment.

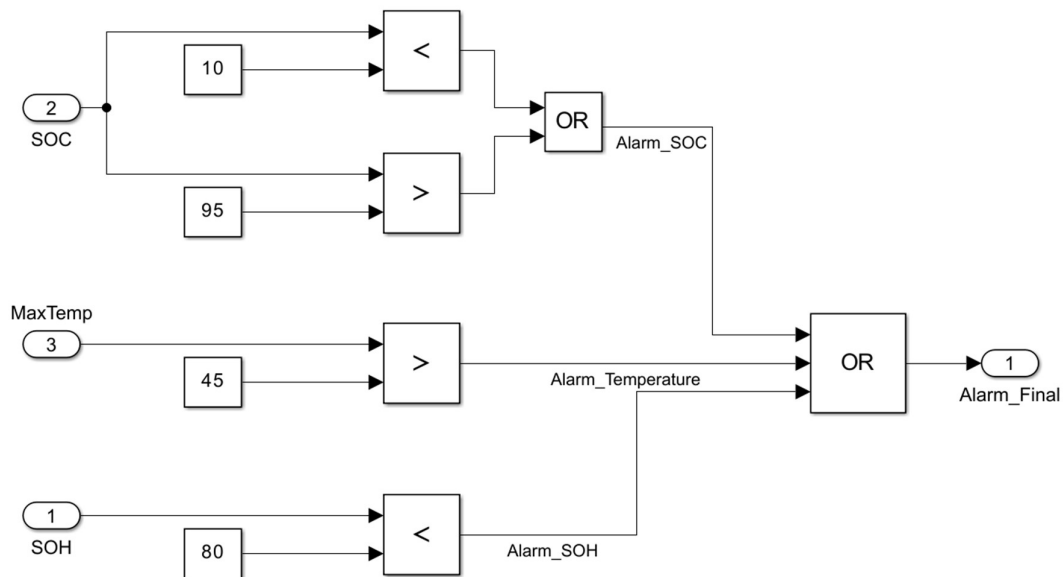


Fig 12: Internal System of Alarm Generator Block

9) DATA COLLECTOR:

The Data Collectors block also captures all relevant input/output data generated during the course of operation. Input/output parameters collected include: SOC, Temperature, Cycle Count, SOH, RUL, and Alarm Status. Data Collector blocks allow operators to collate (i.e., consolidate) all collected data into a single location so that it can easily be stored for future reference and analyzed. When data from the multiple Data Collectors blocks are tracked together, they create a comprehensive view of battery performance over time. Collected datasets may also be visually represented, reported on or used in the development of machine learning algorithms. Also, the structured dataset improves consistency across the multiple Data Collector blocks and allows for ease of sharing and integrating data between multiple external systems. In summary, the Data Collectors block plays an important role by providing an avenue through which simulation results can be converted into useful and actionable information.

10) EXPORT

Finally, the Export block is the final stage of data collection within the system; collected data is then exported to a selected external system. Outputted data is formatted in such a way that allows it to be easily utilized by MATLAB Workspaces, Dashboard applications, and Visualization software. Therefore, all simulation results can be directly incorporated into real-time monitoring systems seamlessly. Through utilizing Export block generated data, operators and researchers will have access to greater knowledge regarding the operation of batteries and may subsequently perform additional analysis and report on results from simulation results, or utilize the exported datasets to create and train machine learning models, or use the datasets to better enable decision-making by utilizing information generated through the simulation. Therefore, the Export block not only closes the loop on the simulation process; it allows for the extension of the simulation process into practical use through the incorporation of simulation results into the external environment. Therefore, the Export block serves as a gateway between simulation results into practical utility for users.

FLOWCHART

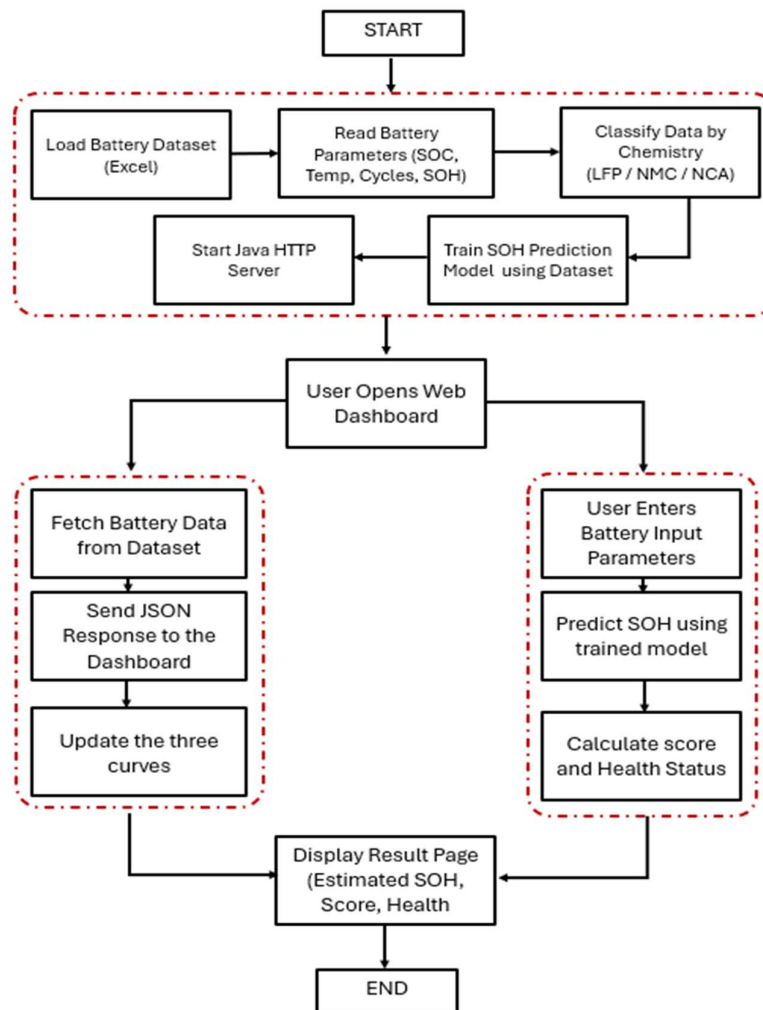


Fig 13: Working of the Entire Project

MATHEMATICAL CALCULATION

Consider the following dataset as a sample used for Random Forest Approach by Boot Strapping Method

Sample	SOC_avg	Temperature (°C)	Cycles	Class
S1	46	18.83	250	Good (1)
S2	46	29.92	249	Good (1)
S3	44	24.91	358	Warning (2)
S4	42	18.96	315	Warning (2)
S5	43	16.13	487	EOL (3)
S6	46	28.51	490	EOL (3)

Now, Consider the first decision tree for SOC_Avg

Sample	SOC_avg	Class
S1	46	Good (1)
S2	46	Good (1)
S3	44	Warning (2)
S4	42	Warning (2)
S5	43	EOL (3)
S6	46	EOL (3)

- 1) Try split between 44 and 46:

Split point:

$$\frac{44 + 46}{2} = 45$$

- 2) Divide the data:

- a. Left Node ($\text{SOC_avg} \leq 45$):

42(2), 43(3), 44(2)

$$\text{Mean}, \bar{y}_L = \frac{2 + 3 + 2}{3} = \frac{7}{3} = 2.33$$

- b. Right Node ($\text{SOC_avg} > 45$):

46(1), 46(1), 46(3)

$$\text{Mean}, \bar{y}_R = \frac{1 + 1 + 3}{3} = \frac{5}{3} = 1.67$$

- 3) Compute MSE:

a. Left Node:

$$\begin{aligned}
 MSE_L &= \frac{(2 - 2.33)^2 + (3 - 2.33)^2 + (2 - 2.33)^2}{3} \\
 &= \frac{(-0.33)^2 + (0.67)^2 + (-0.33)^2}{3} \\
 &= \frac{0.11 + 0.44 + 0.11}{3} = \frac{0.66}{3} = 0.22
 \end{aligned}$$

b. Right Node:

$$\begin{aligned}
 MSE_R &= \frac{(1 - 1.67)^2 + (1 - 1.67)^2 + (3 - 1.67)^2}{3} \\
 &= \frac{(-0.67)^2 + (-0.67)^2 + (1.33)^2}{3} \\
 &= \frac{0.44 + 0.44 + 1.77}{3} = \frac{2.65}{3} = 0.88
 \end{aligned}$$

4) Total MSE

$$MSE = MSE_L + MSE_R = 0.22 + 0.88 = 1.10$$

Now, Consider the second decision tree for Temperature

Sample	Temperature (°C)	Class
S1	18.83	Good (1)
S2	29.92	Good (1)
S3	24.91	Warning (2)
S4	18.96	Warning (2)
S5	16.13	EOL (3)
S6	28.51	EOL (3)

1) Try split between 24.91 and 28.51:

Split point:

$$\frac{24.91 + 28.51}{2} = 26.71$$

2) Divide the data:

a. Left Node (Temperature ≤ 26.71):

18.83(1), 18.96(2), 16.13(3), 24.91(2)

$$\text{Mean, } \bar{y}_L = \frac{1 + 2 + 3 + 2}{4} = \frac{8}{4} = 2$$

b. Right Node (Temperature > 26.71):

29.92(1), 28.51(3)

$$Mean, \bar{y}_R = \frac{1+3}{2} = \frac{4}{2} = 2$$

3) Compute MSE:

a. Left Node:

$$\begin{aligned}MSE_L &= \frac{(1-2)^2 + (2-2)^2 + (3-2)^2 + (2-2)^2}{4} \\&= \frac{(-1)^2 + (0)^2 + (1)^2 + (0)^2}{4} \\&= \frac{-1 + 0 + 1 + 0}{4} = \frac{0}{4} = 0\end{aligned}$$

b. Right Node:

$$\begin{aligned}MSE_R &= \frac{(1-2)^2 + (3-2)^2}{2} \\&= \frac{(-1)^2 + (1)^2}{2} \\&= \frac{-1 + 1}{2} = \frac{0}{2} = 0\end{aligned}$$

4) Total MSE

$$MSE = MSE_L + MSE_R = 0 + 0 = 0$$

Now, Consider the third decision tree for Cycles

Sample	Cycles	Class
S1	250	Good (1)
S2	249	Good (1)
S3	358	Warning (2)
S4	315	Warning (2)
S5	487	EOL (3)
S6	490	EOL (3)

1) Try split between 315 and 358:

Split point:

$$\frac{315 + 358}{2} = 336.5$$

2) Divide the data:

a. Left Node (Cycle ≤ 336.5):

250(1), 249(1), 315(2)

$$\text{Mean, } \bar{y}_L = \frac{1 + 1 + 2}{3} = \frac{4}{3} = 1.33$$

b. Right Node (Cycle > 26.71):

$$358(2), 487(3), 490(3)$$

$$\text{Mean, } \bar{y}_R = \frac{2 + 3 + 3}{3} = \frac{8}{3} = 2.33$$

3) Compute MSE:

a. Left Node:

$$\begin{aligned} \text{MSE}_L &= \frac{(1 - 1.33)^2 + (1 - 1.33)^2 + (2 - 1.33)^2}{3} \\ &= \frac{(-0.33)^2 + (-0.33)^2 + (0.67)^2}{3} \\ &= \frac{0.11 + 0.11 + 0.44}{3} = \frac{0.66}{3} = 0.22 \end{aligned}$$

b. Right Node:

$$\begin{aligned} \text{MSE}_R &= \frac{(2 - 2.33)^2 + (3 - 2.33)^2 + (3 - 2.33)^2}{2} \\ &= \frac{(-0.33)^2 + (0.67)^2 + (0.67)^2}{2} \\ &= \frac{0.11 + 0.44 + 0.44}{2} = \frac{0.99}{2} = 0.33 \end{aligned}$$

4) Total MSE

$$\text{MSE} = \text{MSE}_L + \text{MSE}_R = 0.22 + 0.33 = 0.55$$

Hence, The final decision tree is given by

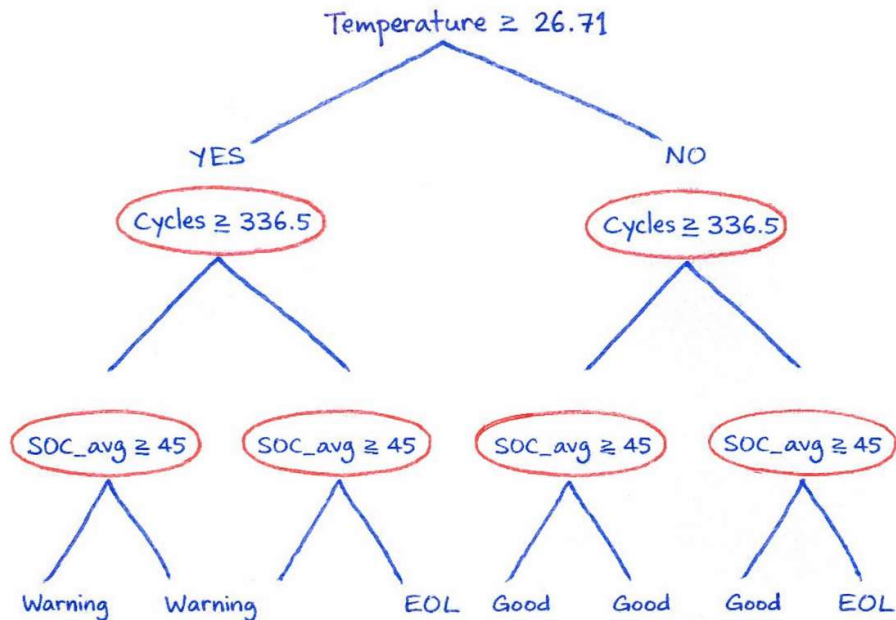


Fig 14: Final Decision tree of the Project

CODING

1. To Predict the Health based on SOH:

```
function y = fcn(SOH)

if SOH < 70
    y = 3;      % EOL
elseif SOH < 80
    y = 2;      % Warning
else
    y = 1;      % Good
end

end
```

2. To Generate 25000 Sample of Dataset from MATLAB

```
t = (0:1:25000)';
chem = randi([1 3], length(t), 1);
chem_profile = [t chem];
```

3. To Convert the Generated Dataset into an Excel File

```
data = out.evbattery;
V = data.signals.values;
T = array2table(V, ...
    'VariableNames', { ...
    'Chemistry_ID', ...
    'SOC_Min_Low', ...
    'SOC_Max_High', ...
    'AvgTemp', ...
    'MaxTemp', ...
    'Cycles', ...
    'SOH', ...
    'Health_Status_Code', ...
    'RUL', ...
    'Alarm_Final'});
Chemistry_Name = strings(height(T),1);
Chemistry_Name(T.Chemistry_ID == 1) = "LFP";
Chemistry_Name(T.Chemistry_ID == 2) = "NMC";
Chemistry_Name(T.Chemistry_ID == 3) = "NCA";
T = addvars(T, Chemistry_Name, 'After', 'Chemistry_ID');
Health_Status = strings(height(T),1);
Health_Status(T.Health_Status_Code == 1) = "Good";
Health_Status(T.Health_Status_Code == 2) = "Warning";
Health_Status(T.Health_Status_Code == 3) = "EOL";
T = addvars(T, Health_Status, 'After', 'Health_Status_Code');
head(T)
writetable(T, 'Battery_Dataset.xlsx');
```

ELECTRIC VEHICLE BATTERY ANALYTICS DASHBOARD

This comprehensive web-based solution provides real-time analysis and monitoring of electric vehicle (EV) batteries. The dashboard provides a very clean, easy-to-use interface that allows users to quickly see and compare multiple types of battery chemistries (e.g., Lithium Iron Phosphate (LFP), Nickel Manganese Cobalt (NMC), Nickel Cobalt Aluminium (NCA)). Each type of battery will be displayed on its own panel to make it easy to read and compare.

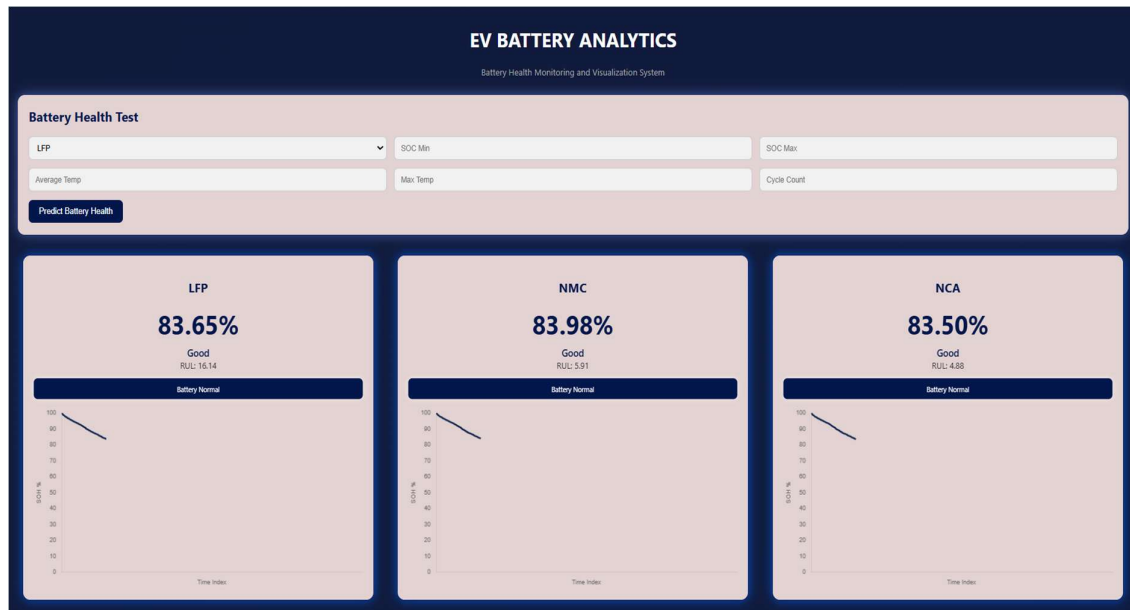


Fig 15: Output Dashboard of the Project

The dashboard displays the following key performance indicators (KPIs):

- State of Health (SOH) – the current condition of the battery as a percentage of its full state of health.
- Remaining Useful Life (RUL) – the estimated time that the battery will continue to operate at optimal levels.

These metrics are essential for predicting when a battery will fail and managing operations around that.

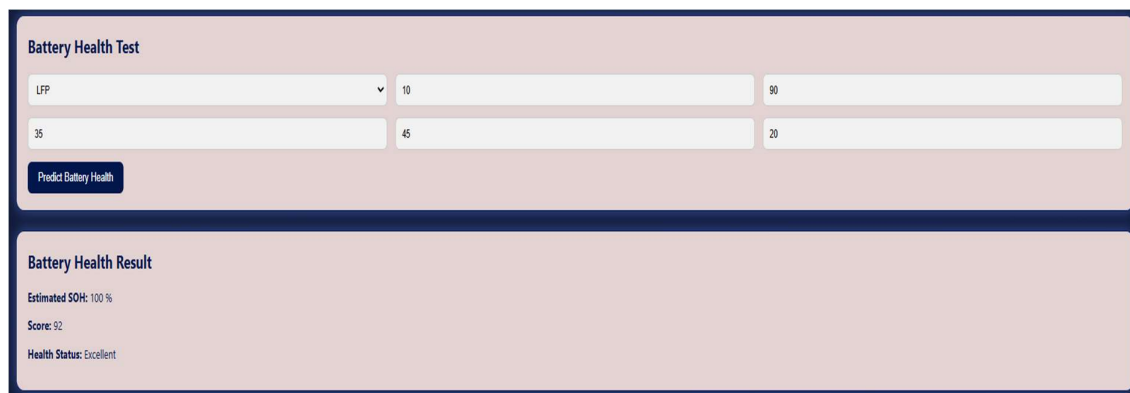


Fig 16: Output SOH Calculator of the Project

The dashboard also provides a quick way to identify battery status, e.g., “Good” or “Battery Normal,” enabling users to quickly understand the battery performance without requiring deep technical analyses. The dashboard includes graphical representations of the SOH for time-series analysis to provide insight into trends of battery degradation. By identifying patterns of degradation, users can take preventive measures and anticipate failures before they occur. The dashboard continuously updates in real-time to ensure your organization has access to the most up-to-date information at all times.

The combination of these advantages allows EV battery monitoring systems to convert complex battery data into actionable information that improves decision-making, better manages battery lifecycles, and increases the reliability and efficiency of EVs through advanced monitoring capabilities and visualization tools. The main advantage is the use of sophisticated battery degradation models to accurately forecast future battery health based on specific characteristics of that battery. Forecasting is based on parameters such as age, charging rates, temperatures and load factors and other common characteristics of a particular battery type. Real-time battery monitoring is one of the other main advantages. Continuous monitoring of temperature, charging cycles, usage patterns, etc. of a battery allows for quick identification of any irregularities and provides an opportunity to prevent them from becoming serious problems. This capability is particularly important in electric vehicle applications where dependability is essential.

The system can also be used with a wide variety of battery chemistries (LFP, NMC, and NCA) and can therefore be used in numerous electric vehicle applications including those for private and commercial use, as well as for the purpose of energy storage. The system allows for preventive maintenance to occur by detecting any early indicators of degradation which will provide the user with a chance to take action before the battery has completely degraded. From a cost-effective standpoint, the use of this system will help lower maintenance expenses and increase the useful life of batteries by helping you better monitor how the battery is being used. By reducing your surprise failures and optimizing how a battery is operated, this system can provide for productivity improvement. By combining predictive analytics with continuous monitoring, you will have a comprehensive solution to address modern battery monitoring requirements and provide for both technical and financial reliability.

RESULT

This project demonstrates the creation of a smart and data-driven system to perform both monitoring and forecasting for electric vehicle battery health. By tracking various input factors (such as SOC, temp, cycles and time used) it was possible to make educated guesses about how and when a battery would become defective. In order to reproduce battery performance in real-time using Simulink, the first step is to generate simulated data based on the battery operating region and any battery characteristics including manufacturer ratings. The second step is to use this data to create a dataset that can be used to compare future data to earlier datasets, thereby ensuring that all future data is consistent with previous results. Next, all processed data is visualized using MATLAB in a structured format for easy analysis and for use in potential future machine learning projects. Such a structured approach allows for future improvement and predictive modelling of battery performance. As a result of this structured approach, the processed data is presented on a dynamic web-based dashboard allowing users to check in with the overall battery status in real-time. Each battery type is represented in the dashboard with its numerical

and graphical data being displayed; additionally, degradation trends can be observed over time and battery health can be assessed via visual indicators.

The system is capable of providing continuous battery performance monitoring and to deliver valid predictions of future battery performance based on historical battery performance. This allows for better planning of battery maintenance and to maximize battery performance, and to promote continued technology development regarding electric vehicles. The delivered results confirmed that the system is able to integrate simulation, data analytics, and real-time data visualization into a cohesive and usable solution.

The data generated by the system provides evidence of improved power factor correction through the use of Static Var Compensators (SVCs). The original system was operating at a power factor of 0.7071 due to significant reactive power usage and therefore, inefficient electrical energy use, which causes additional loss and cost of energy. Static Var Compensators (SVC) have been shown to greatly improve the power factor by making it much closer to 1, at about 0.9976. The improved power factor means that the system has a very low amount of reactive power, which translates into more efficient use of electrical energy. When the power factor is improved, it reduces transmission losses and improves overall performance of the system. There is an obvious economic impact associated with this improvement, as the company is experiencing significant reductions in the amount of electricity it pays, with estimated savings of approximately Rs. 2.02 Lakhs/month. This demonstrates the financial advantages of not only improving power factor but also enhancing technical performance.

The results show that SVCs are extremely effective at optimizing the use of power, while at the same time, greatly reducing energy loss and increasing efficiency within the system. Therefore, SVCs are a valuable solution for industries that operate on a continuous basis and require both stable and efficient power solutions.

CONCLUSION

SVCs are an integral part of the overall design and operation of the modern power system, as they provide dynamic control of reactive power, thereby allowing for better utilization of electrical energy. As such, SVCs play a critical role in reducing reactive power flows, thereby minimizing transmission losses and improving the overall efficiency of power delivery.

In addition to performing the functions of dynamically regulating reactive power and improving the power factor, SVCs provide voltage stability, which is critical to the successful operation of today's electric power systems. The ability of SVCs to provide voltage stabilization under varying load conditions is critical to maintaining consistent voltage levels, thus minimizing voltage fluctuation and reducing the risk of damage to electrical equipment.

SVCs also add to the overall stability of the power system by providing rapid response to disturbances and decreasing the likelihood of power oscillations. Thus, SVCs contribute to the overall security and reliability of electric power systems, particularly in large industrial operations with increasing power demands. The use of SVCs is also increasing in advanced systems, including High Voltage Direct Current (HVDC) transmission systems and industries utilizing Thyristor Controlled Drives (TCD).

From an economic standpoint, SVCs provide reduced electricity costs through improved power factor and avoidance of penalties from utilities, as well as long-term energy savings and improved operational efficiency. In summary, SVCs provide a strong value proposition through both technical and economic advantages. By increasing the efficiency, stability and cost-effectiveness of electric power systems, SVCs are an important part of modern electric power systems and industrial electric power systems.